

LESSON 8.8

SOLUTIONS TO SYSTEMS OF EQUATIONS



LESSON INTRODUCTION

MA.8.AR.4.1 Given a system of two linear equations and specified set of possible solutions, determine which ordered pairs satisfy the system of linear equations.

MA.8.AR.4.2 Given a system of two linear equations presented graphically on the same coordinate plane, determine whether there is one solution, no solution, or infinitely many solutions.

LEARNING TARGETS

- I can determine algebraically if an ordered pair satisfies the given system of linear equations.
- I can verify graphically when a system of linear equations has one solution, no solution, or infinitely many solutions.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

ESSENTIAL QUESTIONS

- How many solutions do systems of equations have?
- How can you determine the number of solutions a system of equations has?

LESSON NARRATIVE

This lesson focuses on solidifying students' understanding of what systems of equations and their solutions are. Students delve deeper into both algebraic and graphical representations of systems when determining and verifying whether specified ordered pairs satisfy the system. Students make connections between the number of solutions found (one, none, and infinitely many) and their graphical representations. This includes understanding why a system made of parallel lines does not have any solutions and a system of two equivalent equations has infinitely many solutions.

COMPONENT SUMMARY

8.8.1 WARM-UP (~5 MIN.)

Match systems of linear equations to their graphs

8.8.3 GUIDED PRACTICE (10 MIN.)

Verify solutions to systems of linear equations algebraically and find solutions graphically

8.8.5 WRAP-UP (~5 MIN.)

Analyze a student's work determining the solutions to a system of linear equations

8.8.2 GUIDED INSTRUCTION (15-20 MIN.)

Learn how to verify if an ordered pair is a solution to a system

8.8.4 YOUR TURN! (10 MIN.)

Determine if ordered pairs are solutions to a system and verify using the graph

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- System of equations

MATERIALS

- (Optional) Scientific calculator

ADDITIONAL PREPARATION

- (Optional) Set up floor grids for the Guided Instruction component prior to class

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may mix up the x and y values when verifying whether an ordered pair satisfies the equations in the system.
- Students may struggle to understand what it means for there to be infinitely many solutions to a system in terms of the graph.

THINGS TO CONSIDER

- Allowing students to use calculators during the lesson will help students focus on discovery and process rather than performing calculations.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

The structure of the 8.8.1 Warm-Up allows students to match systems of linear equations with their graphs. Consider using the Poll the Class routine to get a quick snapshot of student understanding prior to whole-group discussion. For example, students could have index cards of three different colors or white boards to respond to the questions, "Which system matches the (first/second/third) graph?" After polling the class, select students to share how they determined their selection, which could include selecting students who made an error so the class can use mistakes as a learning opportunity to refine their understanding of concepts. When these graphs are revisited during question 3 of the 8.8.2 Guided Instruction, consider using this routine again for students to respond to the questions, "Which system has (one/no/infinitely many solution(s))?" before completing the table.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students will verify the solution to systems algebraically prior to considering the graph for several problems throughout this lesson. Students are expected to solve systems of equations by graphing. Highlighting connections between the graphical and algebraic representations of the systems helps to build stronger understanding of this concept. Developing this deep understanding through multiple representations with intentional scaffolding will build a stronger foundation for students to learn to solve systems algebraically in Algebra 1 and better understanding of how to determine solutions from graphs in this course.

DIFFERENTIATE INSTRUCTION

Consider using the floor of the classroom as a coordinate grid in the 8.8.2 Guided Instruction. Various lines could be graphed on the floor prior to the start of the lesson and during this component, students can walk the lines to model what it means in terms of the graph for a system of two equations to have one, no, or infinitely many solutions. This will help to develop students' conceptual understanding.

8.8.1 WARM-UP: BELL WORK

TEACHER MOVES

Poll the Class

Consider using the "Poll the Class" language routine to get initial responses after students have completed the Warm-Up prior to whole-class discussion. An easy way for students to share their answers is by using mini whiteboards, index cards with letters on them, or colors associated with each system.

ENRICHMENT

If time allows for a whole-class debrief, consider asking the following: How did you determine the matches of the systems and their graphs? (Some students may have matched each equation to a line based on their learning from Unit 7. Others may have used the structure of the equations based on the learning in Lesson 7.)

Some students may connect what they learned yesterday to this Warm-Up and feel comfortable talking about the number of solutions for each system. It is not the expectation they are there yet, though, as these concepts will be solidified formally in the next component.



8.8.1 Warm-Up: Bell Work

Three systems of equations are shown.

System A	System B	System C
$\begin{cases} y = \frac{1}{4}x + \frac{5}{4} \\ y = 1.25 + 0.25x \end{cases}$	$\begin{cases} y = \frac{2}{3}x + 5 \\ y = \frac{1}{3}x + 3 \end{cases}$	$\begin{cases} y = -\frac{1}{6}x + 1 \\ y = -\frac{1}{6}x - 1 \end{cases}$

1. The graphs of each system are shown. Match each system of linear equations with its graphical representation.

Graph			
System	B	C	A



8.8.2 Guided Instruction: Solutions to Systems of Equations

1. Consider the system of equations $\begin{cases} y = \frac{1}{2}x + 5 \\ y = -\frac{1}{4}x + 8 \end{cases}$.

- a. Determine algebraically, using substitution, if the ordered pair $(2, 7.5)$ satisfies the system of linear equations.

Does $(2, 7.5)$ satisfy the equation $y = \frac{1}{2}x + 5$?	Does $(2, 7.5)$ satisfy the equation $y = -\frac{1}{4}x + 8$?
$y = \frac{1}{2}x + 5$	$y = -\frac{1}{4}x + 8$
$7.5 = (\frac{1}{2} \cdot 2) + 5$	$7.5 = (-\frac{1}{4} \cdot 2) + 8$
$7.5 = 1 + 5$	$7.5 = -\frac{1}{2} + 8$
$7.5 = 6$	$7.5 = 7.5$

- b. Complete the statement.

The ordered pair $(2, 7.5)$ ☐ is ☒ is not a solution to the

system. It ☒ does ☐ does not satisfy the equation

$y = -\frac{1}{4}x + 8$, and it ☐ does ☒ does not satisfy the equation

$y = \frac{1}{2}x + 5$. To be a solution to the system, the ordered

pair

- ☐ only needs to satisfy one of the equations.
☒ must satisfy all equations in the system.

- c. Complete the table to determine if each ordered pair satisfies both equations in the system.

Ordered Pair	$y = \frac{1}{2}x + 5$	$y = -\frac{1}{4}x + 8$	Are Both Equations True?
$(-2, 4)$	$4 = (\frac{1}{2} \cdot -2) + 5$ $4 = -1 + 5$ $4 = 4$	$4 = (-\frac{1}{4} \cdot -2) + 8$ $4 = \frac{1}{2} + 8$ $4 \neq 8.5$	☐ Yes ☒ No
$(1, 8)$	$8 = (\frac{1}{2} \cdot 1) + 5$ $8 = \frac{1}{2} + 5$ $8 \neq 5.5$	$8 = (-\frac{1}{4} \cdot 1) + 8$ $8 = -\frac{1}{4} + 8$ $8 \neq 7.75$	☐ Yes ☒ No
$(4, 7)$	$7 = (\frac{1}{2} \cdot 4) + 5$ $7 = 2 + 5$ $7 = 7$	$7 = (-\frac{1}{4} \cdot 4) + 8$ $7 = -1 + 8$ $7 = 7$	☐ Yes ☒ No

- d. What is the solution to the system $\begin{cases} y = \frac{1}{2}x + 5 \\ y = -\frac{1}{4}x + 8 \end{cases}$?
- $(4, 7)$

8.8.2 GUIDED INSTRUCTION: SOLUTIONS TO SYSTEMS OF EQUATIONS

TEACHER MOVES

Lead students through the guided instruction using whole-group instruction. Use teacher Think-Aloud strategies to model question 1 and then for the remainder of the component.

DIFFERENTIATE INSTRUCTION

Consider using different colors when substituting the x and y values while modeling how to determine if an ordered pair satisfies the given system. This allows students to better follow the work visually, while Think-Aloud strategies are used to share the logic behind the computation.

TEACHER MOVES

Think-Pair-Share

Model how to complete the table for the first ordered pair. Then consider utilizing the Think-Pair-Share language routine for the remaining two ordered pairs to provide students with the opportunity to analyze independently and then discuss with a partner prior to sharing as a whole group.

TEACHER MOVES

Consider giving students an opportunity to discuss why $(4, 7)$ is a solution to the system using the definition from Lesson 7. Listen as students discuss to identify student thinking to highlight or to clarify misconceptions.

Students should recognize that this is the solution because when the x - and y -values are substituted into both equations, both equations are true.

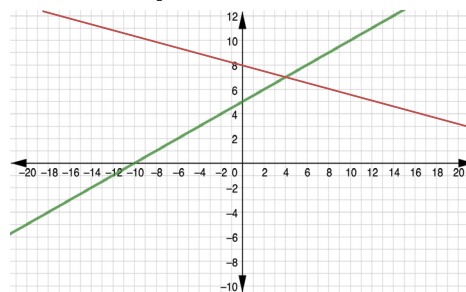
8.8.2 GUIDED INSTRUCTION: SOLUTIONS TO SYSTEMS OF EQUATIONS (CONTINUED)

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

Students verified the solution to this system algebraically prior to considering the graph. Students will solve systems of equations by graphing, so expressing connections between the graphical and algebraic representation of the systems helps to build stronger understanding of this concept. Developing this deep understanding through multiple representations with intentional scaffolding will build a stronger foundation for students to learn to solve systems algebraically in Algebra 1.

2. The graph of $y = \frac{1}{2}x + 5$ is shown on the coordinate grid.



- a. Graph $y = -\frac{1}{4}x + 8$ on the same coordinate grid.
Shown on coordinate grid.
- b. Describe how the solution to the system $\begin{cases} y = \frac{1}{2}x + 5 \\ y = -\frac{1}{4}x + 8 \end{cases}$ determined in Part D of the previous question relates to the graph of the system.
Answers will vary. The graph represents the same system of equations. The solution to the system is where the two equations intersect on the graph. They intersect at (4,7), which is the same solution verified in Part D of the previous question.
- c. How many solutions are there for this system?
one

TEACHER MOVES

Help students make the connection to solving one-variable equations when there can be one solution, infinitely many solutions, or no solutions. Question 3 creates the opportunity for students to formalize the concepts they explored in Lesson 7. Provide appropriate guidance through this problem to ensure understanding.

Consider revisiting the Poll the Class routine from the warm-up to ask students about which systems of one, no, or infinitely many solutions prior to completing the table.

When completing the table, make connections between the statements in the last column and the graphs and equations to ensure students are making the necessary connections between concepts.

Systems of equations in two variables can have one solution, no solution, or infinitely many solutions.

3. The systems of equations and their corresponding graphs from the Warm-Up are shown again in the table below.
- a. Complete the description of the solution(s) for each system using the phrase **one solution**, **no solution**, or **infinitely many solutions**.

	System	Graph	Solution(s)
A	$\begin{cases} y = \frac{1}{4}x + \frac{5}{4} \\ y = 1.25 + 0.25x \end{cases}$		This system has <i>infinitely many solutions</i> because the equations are equivalent. Every ordered pair on both lines is the same.
B	$\begin{cases} y = \frac{2}{3}x + 5 \\ y = \frac{1}{3}x + 3 \end{cases}$		This system has <i>one solution</i> because there is only one ordered pair that is on both lines. It is the point of intersection.
C	$\begin{cases} y = -\frac{1}{6}x + 1 \\ y = -\frac{1}{6}x - 1 \end{cases}$		This system has <i>no solution</i> because there is no ordered pair that is on both lines. There is no point of intersection.

b. Complete the statements.

System

☐ A
☐ B
☐ C

 has no solution because the lines in this system are parallel. Parallel lines have the same

☐ slope,
☐ y-intercept,

 so there

☐ is no point
☐ is one point
☐ are infinitely many points

 where the lines will intersect.

8.8.2 GUIDED INSTRUCTION: SOLUTIONS TO SYSTEMS OF EQUATIONS (CONTINUED)

TEACHER MOVES

Allow students to consider the statements and discuss with a partner prior to going over this with the class to ensure understanding and answer questions before going to the next component.

DIFFERENTIATE INSTRUCTION

Consider creating a coordinate grid on the floor and allowing two students to walk graphed lines for each type of system. This will create an additional entry point for kinesthetic learners regarding when a system has one, no, or infinitely many solutions.

8.8.3 GUIDED PRACTICE: SOLUTIONS TO SYSTEMS OF EQUATIONS

DIFFERENTIATE INSTRUCTION

Consider positioning students to work together by turning question 1a into a jigsaw activity where different groups of students are responsible for the different ordered pairs being considered. Once students finish evaluating their given ordered pair, they then share their answer with the other students responsible for the other ordered pairs and discuss what this means in terms of solutions to the system.

TEACHER MOVES

Consider allowing students to use a scientific calculator to verify whether each ordered pair satisfies the system. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator.

ENRICHMENT

Consider asking the following probing questions, if needed:

- Do you think a system of linear equations can have only two solutions instead of infinitely many solutions? Why or why not?
- Can you conclude that a system of linear equations has an infinite number of solutions if you find more than one ordered pair that satisfies the system? Why or why not?



8.8.3 Guided Practice: Solutions to Systems of Equations

1. Consider the system of equations $\begin{cases} y = \frac{2}{5}x + \frac{6}{5} \\ y = 1.2 + 0.4x \end{cases}$.

- a. Complete the table to determine if each ordered pair satisfies both equations in the system.

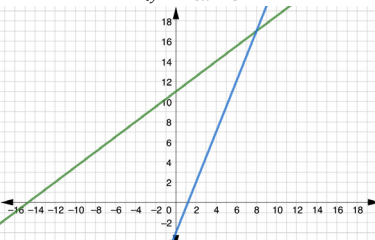
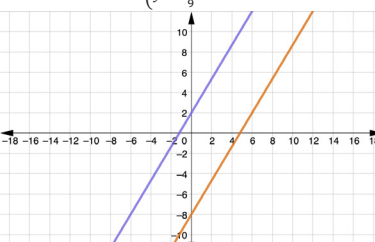
Ordered Pair	$y = \frac{2}{5}x + \frac{6}{5}$	$y = 1.2 + 0.4x$	Are Both Equations True?
$(-1.5, 0.6)$	$0.6 = (\frac{2}{5} \cdot -1.5) + \frac{6}{5}$ $0.6 = -0.6 + 1.2$ $0.6 = 0.6$	$0.6 = 1.2 + (0.4 \cdot -1.5)$ $0.6 = 1.2 + (-0.6)$ $0.6 = 0.6$	☐ Yes ☐ No
$(0, 1.2)$	$1.2 = (\frac{2}{5} \cdot 0) + \frac{6}{5}$ $1.2 = 0 + 1.2$ $1.2 = 1.2$	$1.2 = 1.2 + (0.4 \cdot 0)$ $1.2 = 1.2 + 0$ $1.2 = 1.2$	☐ Yes ☐ No
$(2, 2)$	$2 = (\frac{2}{5} \cdot 2) + \frac{6}{5}$ $2 = \frac{4}{5} + \frac{6}{5}$ $2 = \frac{10}{5}$ $2 = 2$	$2 = 1.2 + (0.4 \cdot 2)$ $2 = 1.2 + 0.8$ $2 = 2$	☐ Yes ☐ No

- b. Based on your work in Part A, complete the statement.

The system $\begin{cases} y = \frac{2}{5}x + \frac{6}{5} \\ y = 1.2 + 0.4x \end{cases}$ has

- ☐ one solution,
☐ no solutions.
☒ infinitely many solutions.

2. For each system of equations, determine whether there is one solution, no solution, or infinitely many solutions. If there is only one solution, give the solution.

System of Equations	Solution(s)
$\begin{cases} y = \frac{3}{4}x + 11 \\ y = 2.5x - 3 \end{cases}$ 	one solution (8, 17)
$\begin{cases} y = \frac{5}{3}x - 8 \\ y = \frac{15}{9}x + 2 \end{cases}$ 	no solutions

8.8.3 GUIDED PRACTICE: SOLUTIONS TO SYSTEMS OF EQUATIONS (CONTINUED)

QUESTIONING

Challenge students to verify their solutions algebraically after they determine the solutions graphically for question 2.

8.8.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider your students and their needs when choosing how to best complete this component of the lesson. Consider providing 3 options based on observations from previous lessons:

- Partner work
- Individual assignments
- Whole-group instruction



8.8.4 Your Turn!

1. Consider the system of equations $\begin{cases} y = \frac{5}{6}x - 2 \\ y = 6 - \frac{1}{2}x \end{cases}$.

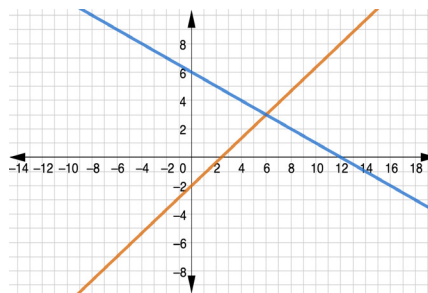
- a. Complete the table to determine if each ordered pair satisfies both equations in the system.

Ordered Pair	$y = \frac{5}{6}x - 2$	$y = 6 - \frac{1}{2}x$	Are Both Equations True?
(4,4)	$4 = \left(\frac{5}{6} \cdot 4\right) - 2$ $4 = \frac{20}{6} - 2$ $4 \neq \frac{8}{6}$	$4 = 6 - \left(\frac{1}{2} \cdot 4\right)$ $4 = 6 - 2$ $4 = 4$	☐ Yes ☒ No
(6,3)	$3 = \left(\frac{5}{6} \cdot 6\right) - 2$ $3 = 5 - 2$ $3 = 3$	$3 = 6 - \left(\frac{1}{2} \cdot 6\right)$ $3 = 6 - 3$ $3 = 3$	☒ Yes ☐ No
(12,0)	$0 = \left(\frac{5}{6} \cdot 12\right) - 2$ $0 = 10 - 2$ $0 \neq 8$	$0 = 6 - \left(\frac{1}{2} \cdot 12\right)$ $0 = 6 - 6$ $0 = 0$	☐ Yes ☒ No

- b. Based on your work in Part A, complete the statement.

The system $\begin{cases} y = \frac{5}{6}x - 2 \\ y = 6 - \frac{1}{2}x \end{cases}$ has ☒ one solution, (6,3).
☐ no solutions.
☐ infinitely many solutions.

- c. The graph of the system $\begin{cases} y = \frac{5}{6}x - 2 \\ y = 6 - \frac{1}{2}x \end{cases}$ is shown.



Use the graph to explain why your answer to Part B is correct.

Answers will vary. This system has one solution because there is only one ordered pair, the point of intersection, that is a solution for both equations.



8.8.5 Wrap-Up: Error Analysis

Bevan solved the system of equations $\begin{cases} y = \frac{3}{4}x - 1 \\ y = \frac{1}{4}x - 3 \end{cases}$.

She used the ordered pairs shown in the table to determine the solution(s) to the system. She put a green checkmark in each of the boxes where the ordered pair satisfied the equation.

Ordered Pair	$y = \frac{3}{4}x - 1$	$y = \frac{1}{4}x - 3$
(2, 0.5)	$0.5 = \frac{3}{4}(2) - 1 = \frac{3}{2} - 1$ $0.5 = \frac{1}{2}$ ✓	$0.5 = \frac{1}{4}(2) - 3 = \frac{1}{2} - 3$ $0.5 \neq -\frac{7}{2}$
(0, -3)	$-3 = \frac{3}{4}(0) - 1 = 0 - 1$ $-3 \neq -1$	$-3 = \frac{1}{4}(0) - 3 = 0 - 3$ $-3 = -3$ ✓
(-4, -4)	$-4 = \frac{3}{4}(-4) - 1 = -3 - 1$ $-4 = -4$ ✓	$-4 = \frac{1}{4}(-4) - 3 = -1 - 3$ $-4 = -4$ ✓

She concluded there are infinitely many solutions to the system because more than one point satisfies the equations.

- Write a short note to Bevan explaining the error she made.
Answers will vary. Your error is that all of the points do not satisfy both equations and systems of equations have infinitely many solutions only when every ordered pair on both lines is the same. For example, the point (0, -3) only satisfies one of the equations. It would need to satisfy both to be a solution to the system.

8.8.5 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

Consider giving students the option to make a video rather than writing a note to describe and show the where the error occurred.

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in Lessons 9-10.